

# ASHUTOSH BARI

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## PROFESSIONAL SUMMARY

AI Engineer specializing in real-time computer vision systems and production ML deployment. Hands-on experience building live surveillance and traffic monitoring systems using YOLOv8 at FUTOPS TECHNOLOGIES. Expertise in end-to-end ML pipelines, real-time video stream processing, model optimization, and containerized deployment. Proven ability to deliver scalable, production-grade AI solutions for mission-critical applications.

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## TECHNICAL SKILLS

Programming & ML: Python | NumPy | Pandas | Scikit-learn | Matplotlib | Seaborn

Computer Vision & AI: YOLOv8 | OpenCV | Object Detection | Real-time Video Processing | Vehicle Tracking | Motion Analysis

MLOps & Deployment: Docker | Linux (CLI) | Git/GitHub | Flask | Model Optimization | Inference Pipeline Design

Data & Visualization: MySQL | Power BI | Tableau | Advanced Excel | SQL

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## PROFESSIONAL EXPERIENCE

**AI Engineer Intern | FUTOPS TECHNOLOGIES India Pvt. Ltd. | 4 Months**

### Real-Time AI Systems Development:

- Architected and deployed production-grade Gunny Bag Detection system using YOLOv8 for live government surveillance
  - Built end-to-end ML pipeline: custom data annotation, preprocessing, model training, hyperparameter tuning, and real-time inference
  - Optimized YOLOv8 model for challenging real-world conditions including low-light environments and varying traffic densities
  - Achieved high precision and recall metrics through iterative model evaluation using mAP, precision-recall curves, and confusion matrices
  - Deployed containerized solutions using Docker in Linux production environments with robust error handling and logging
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### ATCC – Advanced Traffic Control & Command System:

- Developed real-time Opposite Direction Detection using multi-object tracking and motion vector analysis for live traffic streams
  - Implemented Triple Seat Rider Detection with bounding box analysis and rider count logic, processing 30+ FPS video feeds
  - Built Speed Detection module using frame-based distance-time calculations with calibrated pixel-to-meter conversion
  - Designed No Parking Area Detection using ROI-based monitoring and stationary-time threshold algorithms
  - Optimized inference pipeline for real-time video stream processing, reducing latency and improving detection accuracy
  - Collaborated with cross-functional teams to integrate AI modules into live government surveillance infrastructure
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## PROJECTS

**Real-Time Smoking Detection System | YOLOv8, OpenCV, Streamlit**

- Developed end-to-end computer vision system for smoking detection in video streams and RTSP feeds

- Implemented real-time inference pipeline with frame-by-frame processing, alert generation, and cropped image snapshots
  - Built interactive Streamlit dashboard with live monitoring, alert history, and detection analytics
  - Deployed containerized solution with modular architecture for scalability and maintainability
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#### **Battery Health Prediction** | Scikit-learn, Flask

- Built regression models (Linear Regression, Random Forest) for predictive battery health analysis
  - Evaluated model performance using  $R^2$  score and MSE, achieving high prediction accuracy
  - Developed Flask-based REST API for real-time battery health predictions with user-friendly web interface
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#### **Amazon Sales Analytics** | Power BI, SQL

- Designed interactive Power BI dashboard analyzing sales trends, customer behavior, and revenue patterns
  - Performed complex SQL queries for data extraction, transformation, and aggregation
  - Generated actionable business insights driving data-driven decision-making
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#### **EDUCATION**

Bachelor of Engineering in Electronics & Telecommunication

Savitribai Phule Pune University | 2021 – 2025 | CGPA: 7.2/10

Higher Secondary Certificate (HSC): 85.60%

Secondary School Certificate (SSC): 81.20%

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#### **CERTIFICATIONS**

Machine Learning | Power BI | SQL | Python Programming – ITVEDANT